

ChartVision

Candlestick Pattern Recognition + Grad-CAM Interpretability

30

US tickers

3,417

labelled chart images

5

classes

60.2%

test accuracy

+0.13

macro-F1 vs baseline

MOTIVATION

Why this project?

- 01 Traders read **shapes** on price charts – not the raw numbers.
- 02 So this is a **pure Computer-Vision problem**, not a time-series problem.
- 03 I add **Grad-CAM** so the model can also explain *where* it looked.



The model sees only these images.

Five pattern classes

Head & Shoulders



Reversal

Bearish

Double Top



Reversal

Bearish

Double Bottom



Reversal

Bullish

Bull Flag



Continuation

Bullish

No Pattern



Negative class

Neutral

Head & Shoulders, Double Top, Double Bottom and Bull Flag are the **positive** classes; **No Pattern** is the negative.

End-to-end pipeline: from real prices to a trained model

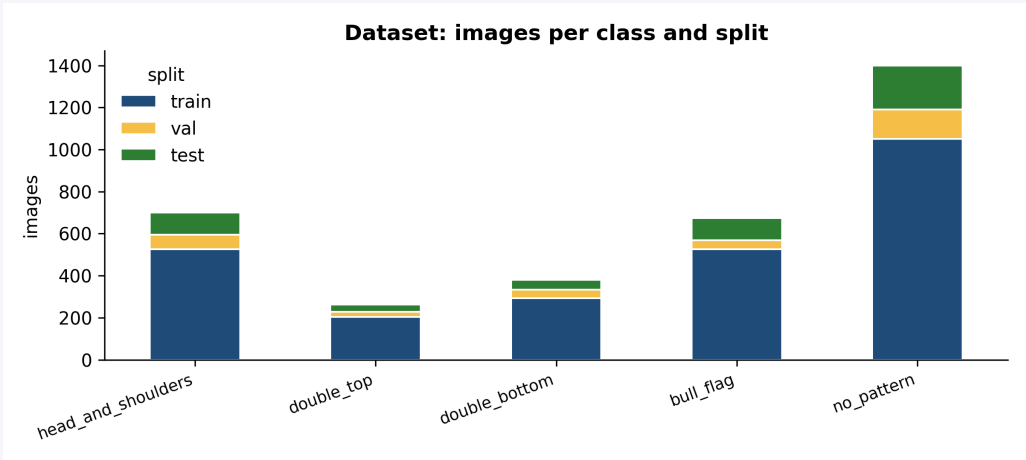


1 YAML config • fixed seed • fully reproducible

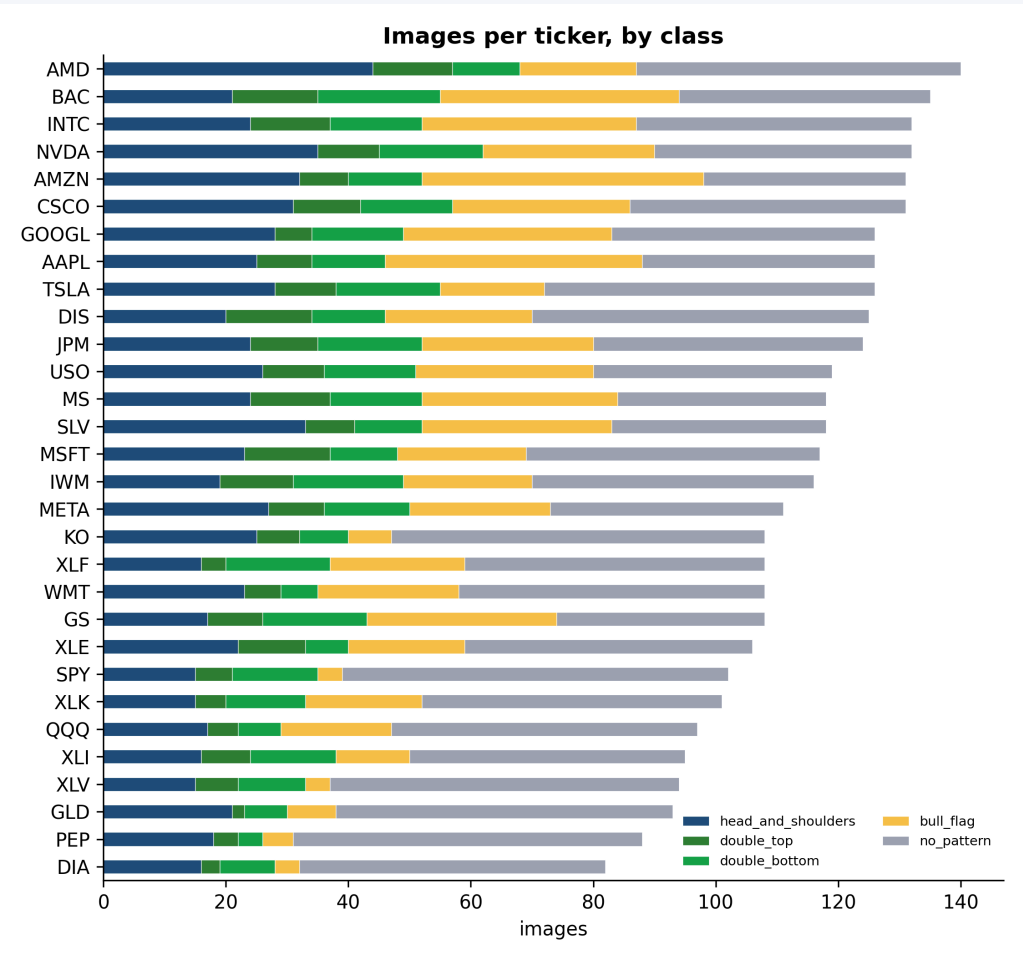
Splits are by date — train < 2022, val 2022, test 2023-24 — so no overlapping windows leak.

THE DATASET

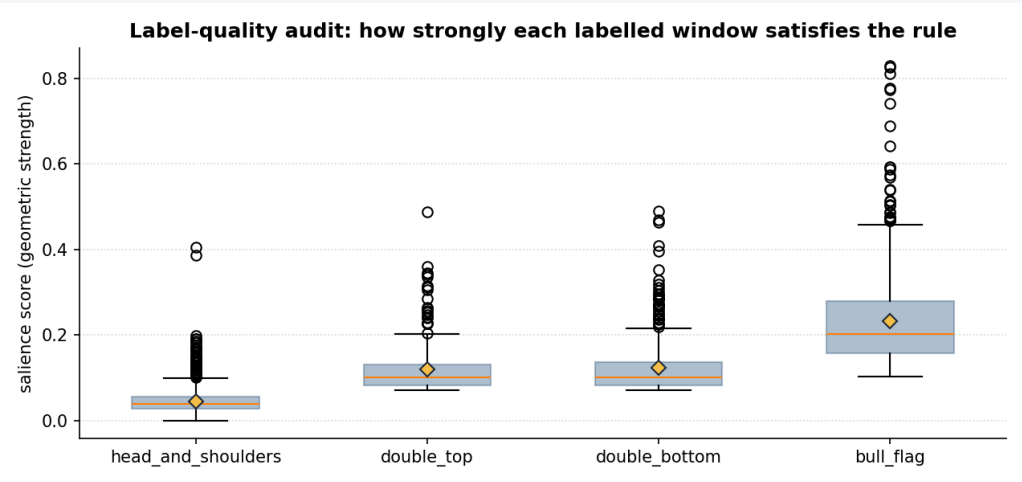
3,417 labelled images, balanced by class and date



Per class & split



Per ticker – all 30 contribute



Label-quality audit per class

● 14 / 14 unit tests passing

Four geometric rules, fully documented and unit-tested

PATTERN	DEFINING RULE
Double Top	2 equal tops ≥ 12 bars apart, trough $\geq 7\%$ deep, drop $\geq 4\%$ after.
Double Bottom	Mirror of Double Top.
Head & Shoulders	Head $\geq 3\%$ above both shoulders; shoulders within 3% of each other.
Bull Flag	Pole $\geq 10\%$ rise in ≤ 15 bars, then a tight flag ($\leq 7\%$ range, $\leq 50\%$ retracement).

A window is a positive only if **exactly one** rule fires; ambiguous windows are dropped, and none-firing windows become **no_pattern**.

Two models, compared on identical splits

Baseline CNN

61k params

3 conv blocks + global average pooling + FC.

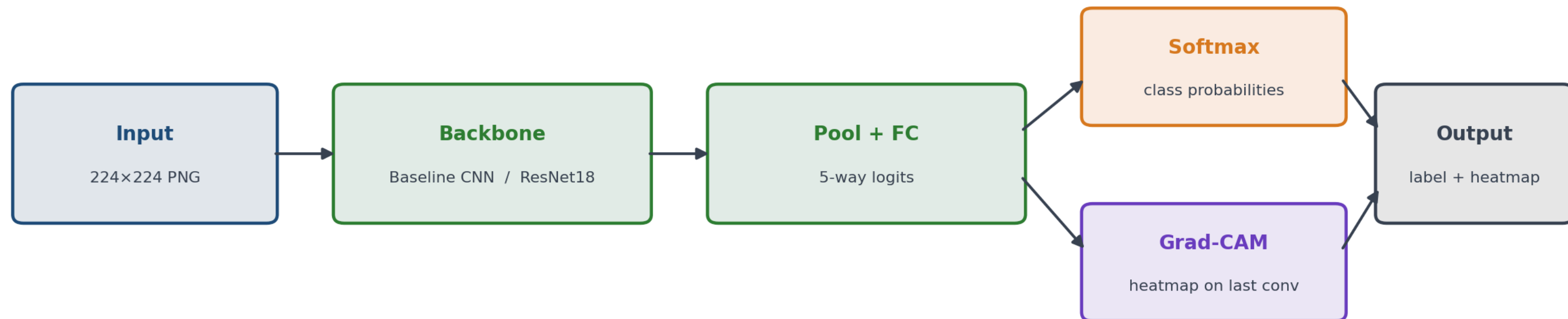
Performance floor.

ResNet18

~8.4M trainable

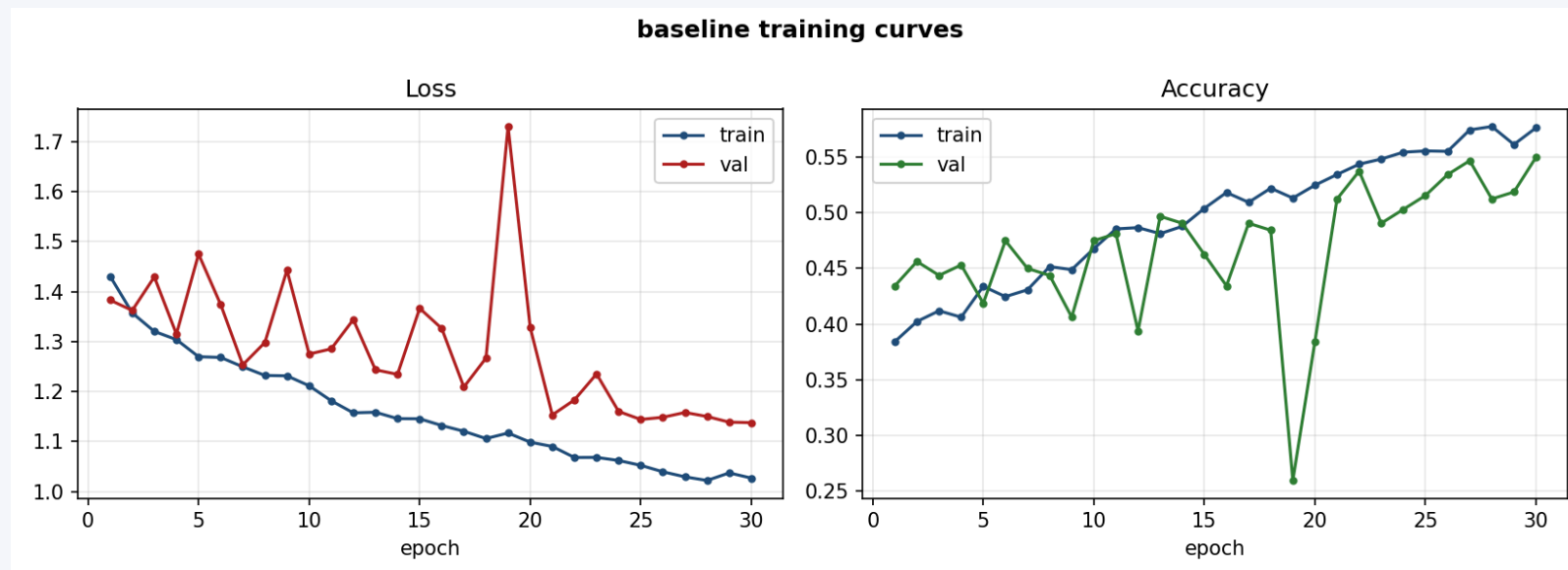
ImageNet pre-trained; layer4 + new FC fine-tuned.

Transfer learning.

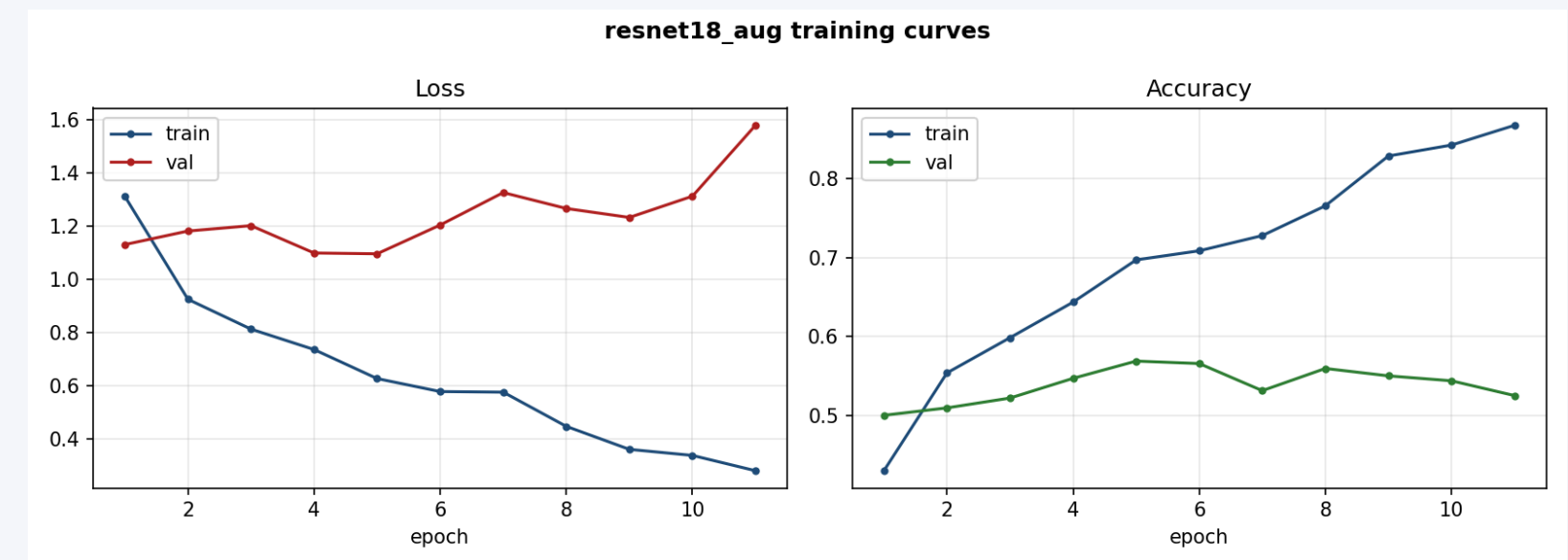


TRAINING BEHAVIOUR

Baseline plateaus; ResNet18 overfits, which motivated the ablation

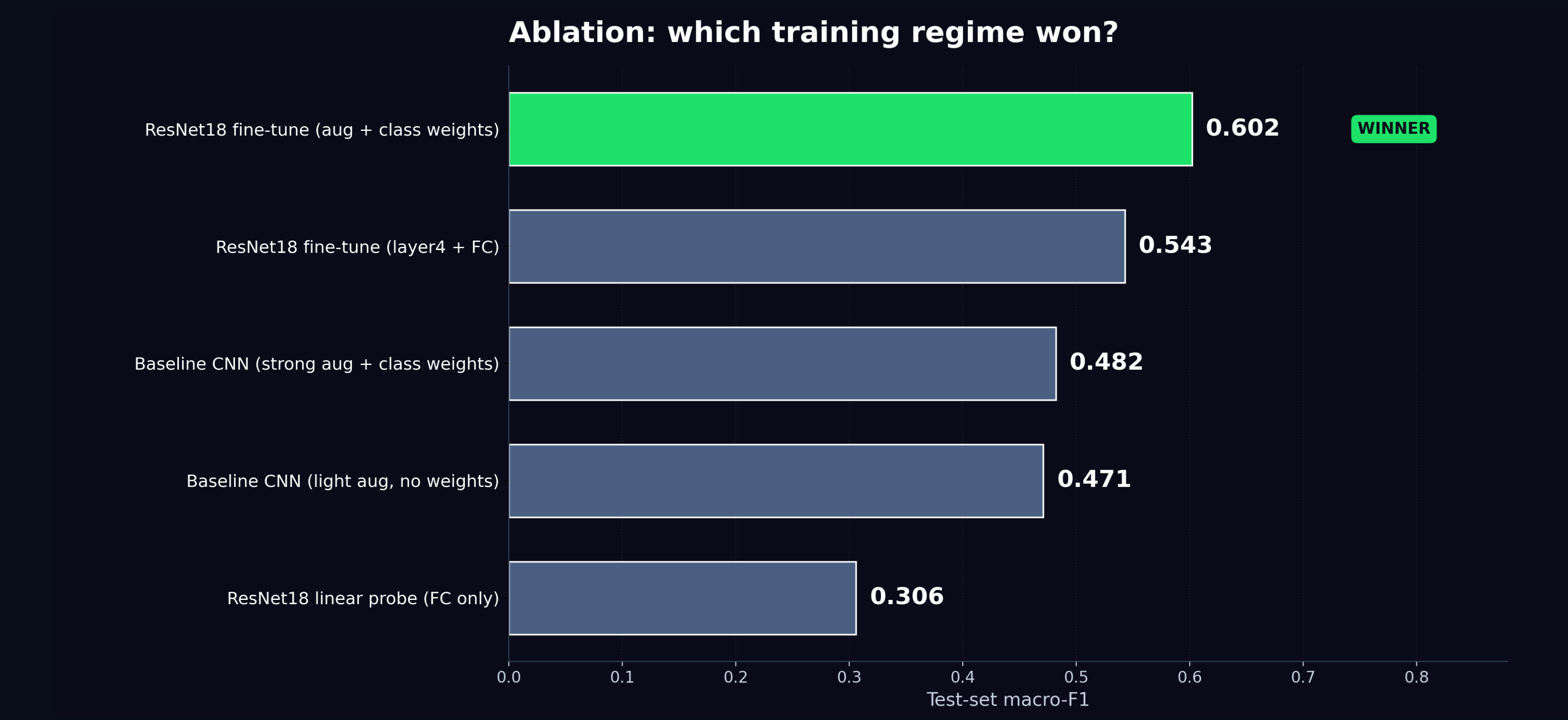


Baseline CNN — loss flattens early.



Validation loss rises after epoch 1 — classic over-fitting. Early stop at epoch 7.

Five training regimes tried – the winner combined three ideas



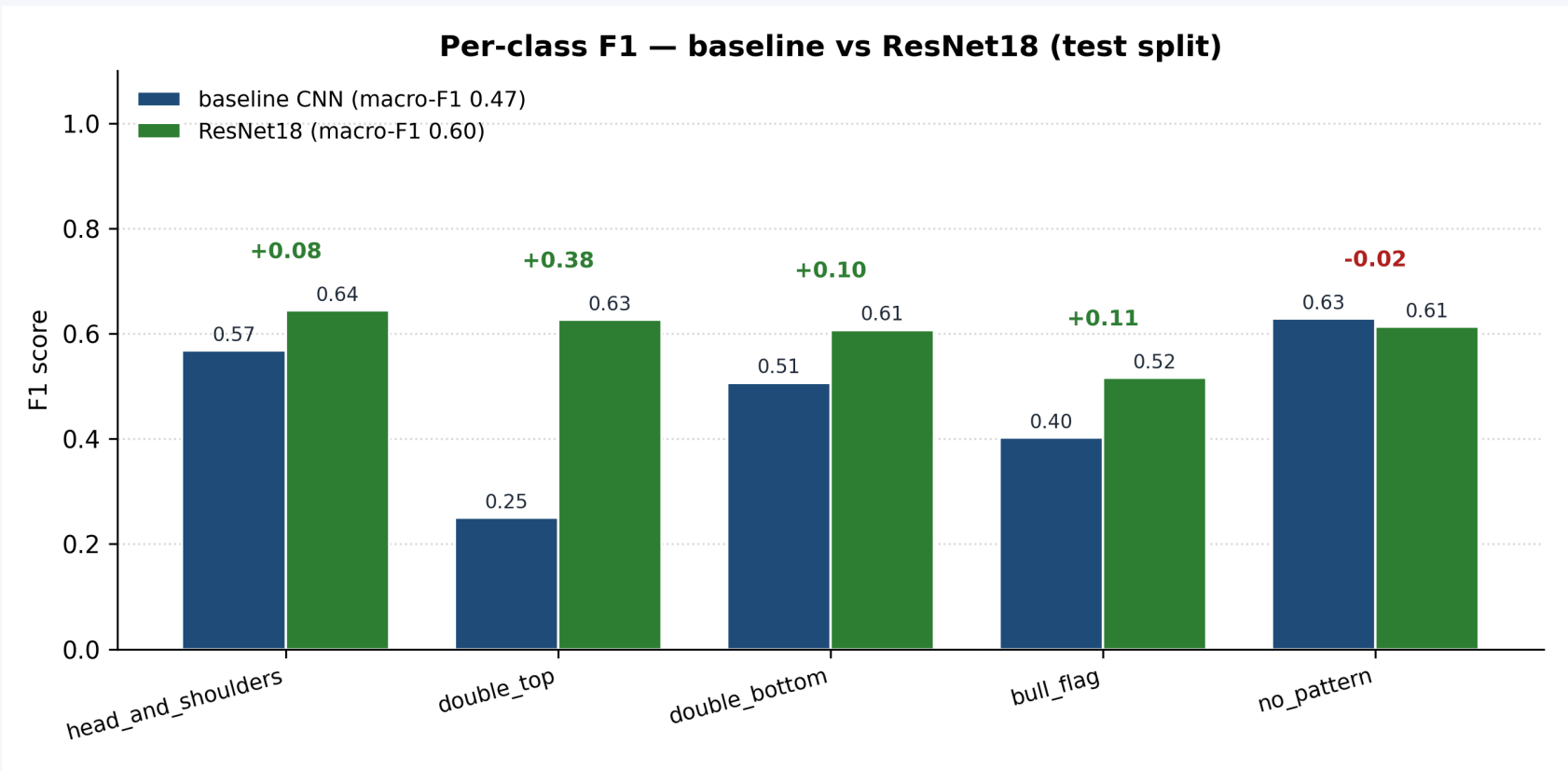
Class weighting + strong augmentation + layer4 fine-tune. Removing any one ingredient drops macro-F1 by ≥ 0.06 .

RESULTS

Test-set results – 502 images, 2023–24, never touched in training

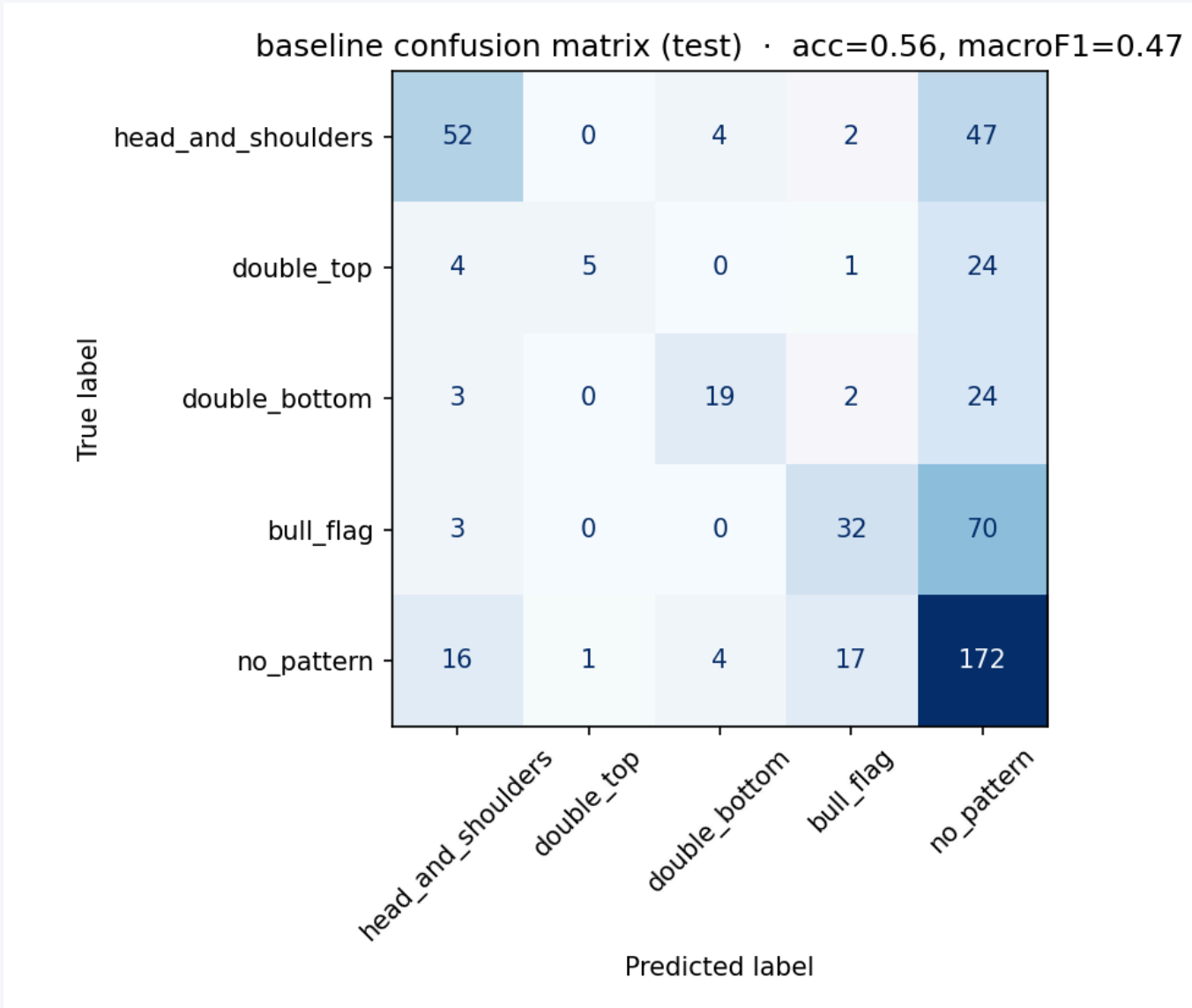
MODEL	ACC	MACRO-F1	ECE
Baseline CNN	55.8%	0.471	0.073
ResNet18 (best)	60.2%	0.602	0.098
Δ	+4.4 pp	+0.131	+0.025

Every positive class improves; the largest jump is **double_top** (F1 0.25 → 0.63, **+0.38**).

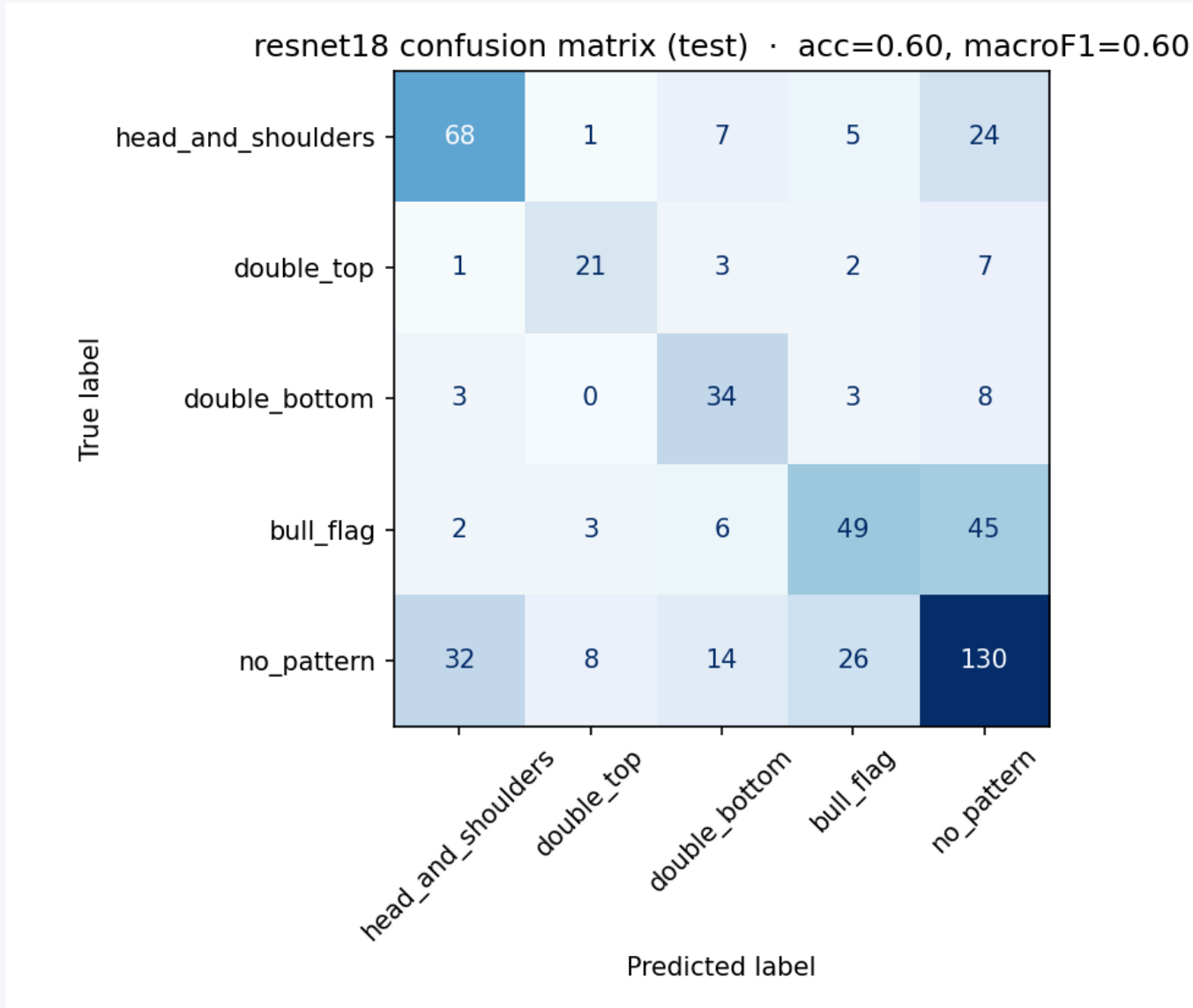


Where the lift came from

Baseline



ResNet18



The baseline (left) dumps the hard classes into **no_pattern**; ResNet18 (right) recovers them.

Grad-CAM — the model looks at the right places

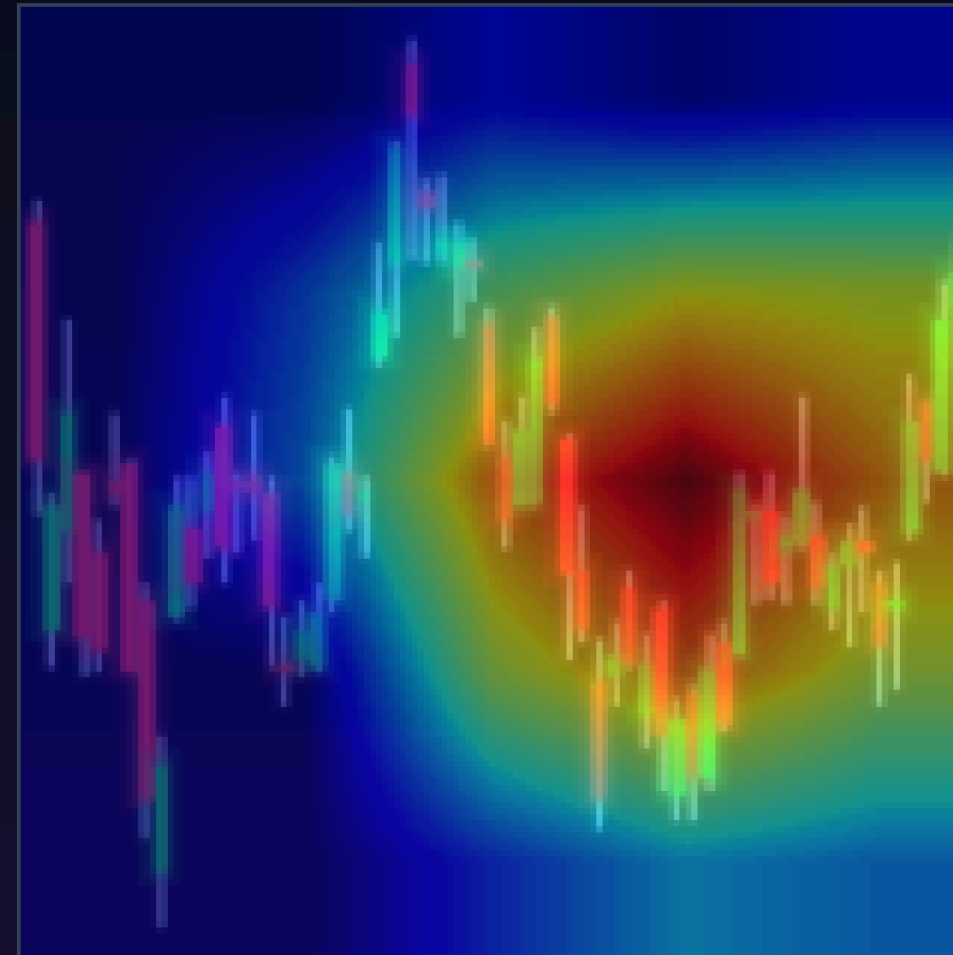
Grad-CAM on a correctly classified Head & Shoulders

ResNet18 prediction confidence 94%

Input chart



Where the model looked



*The heatmap localises on the 'head' (the highest central peak),
validating that the network is using the pattern's defining structure.*

Full per-class grids and a confidently-wrong failure grid are in the report (Figs 9 & 10).

CONCLUSION

What I built, what I'd do next

CONTRIBUTION

- Reproducible CV pipeline (1 config, 1 seed, 4 commands).
- **Macro-F1 0.60** on 5 classes with full per-class balance.
- Grad-CAM correct + failure grids for interpretability.
- Geometric label-quality audit + interactive Plotly demo.

FUTURE WORK

- Manual relabel pass on ~50 charts × 5 classes.
- Bigger backbone (ResNet50, EfficientNet).
- Multi-time-frame data (daily + weekly).
- Hybrid labelling (rules + small hand-labelled set).

Thank you · **Questions?**